

Figure 4: Qualitative results of novel poses synthesis on real data. This novel pose results reflect the accuracy of the reconstructed geometry to a certain extent.

rors has to be used. So, we compare our method with using the SMPL mesh in weights calculation. Besides, using the part-wise design achieves accurate light visibility estimation, which is crucial to generate self-occlusion effects on bodies. To evaluate it, we compare it with two alternatives, removing the light visibility module and using only one neural network to predict the light visibility.

Mesh-based Inverse Skinning. As shown in Tab.2, the reconstruction errors without the mesh-based inverse skinning are consistently larger. We also show the qualitative result in Fig.4, using SMPL mesh to compute the skinning weights leads to artifacts in the contact regions.

Besides, we evaluate the effect of the condition of skinning weights $\mathbf{W}$ (in Sec. 3.1) in the invertible deformation network. As shown in Tab.2, we can also find that removing the condition of $\mathbf{W}$ also leads to worse results.

Part-wise Visibility Estimation. We show quantitative comparisons in Tab.1, the results show that although the albedo map reconstruction qualities are similar, our method with part-wise visibility estimation achieves the best results on relighting. Furthermore, We show qualitative results in Fig.5. We can see that without light visibility modeling (results in the fourth column), the self-occlusion effect cannot be generated at all. With the baseline light visibility modeling, self-occlusion can be partly generated for some poses (results of the third column). For our final solution, the received lighting for different body regions on the novel poses are well modeled and thus the relighting results are consistent with the ground truth rendering.

5 Limitations

The network training takes about 2.5 days in total on a single RTX 3090 GPU, and it takes about 40 seconds to render an image with a resolution of 512×512 during inference (more details in the supplemental document). Integrating instant training techniques like Instant-NGP (?) may improve the efficiency of our technique. It is still hard for our method to animate pose-dependent wrinkle deformations (especially for loose clothing) or generate global illumination effects, which are also open problems in this topic. Our method only considers the body motion rather than the face and hands, while recent works (??) provide possibilities to handle them.

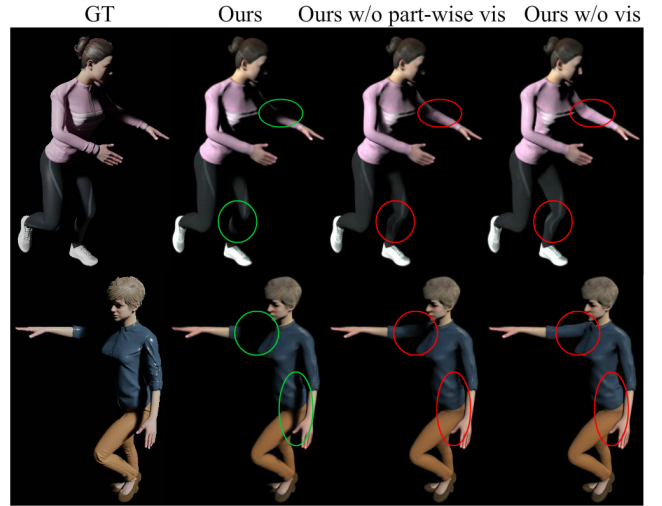


Figure 5: Ablation study on part-wise light visibility. See our method synthesizing plausible self-occlusions.

6 Conclusion

This is the first work that reconstructs relightable and animatable neural avatars with plausible shadow effects from sparse human videos. For dynamic body geometry modeling, the proposed invertible deformation field provides a novel and effective way to solve the inverse skinning problem. Besides, the part-wise light visibility modeling solves the problem of dynamic object relighting based on neural fields. Benefiting from the two techniques, our method succeeds in disentangling the geometry, material of the clothed body, and lighting, thus building a relightable and animatable neural avatar in a lightweight setting.

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